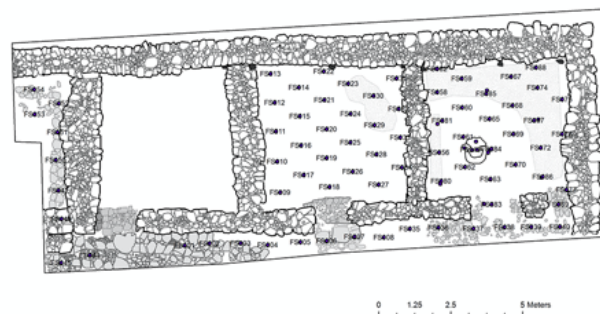


Recovering Plant Remains

Sampling Archaeological Sites

There are several ways to sample an archaeological site for paleoethnobotanical remains. One method, most common for macroremains and phytoliths, involves taking soil samples from various archaeological contexts. Ideally, a sample is taken from every unit of soil identified as stratigraphically distinct during excavation. This includes layers related to construction or filling of structures, as well as samples from features where people are likely to have used or discarded plant materials, including hearths, ovens, storage areas, and trash deposits (middens). This type of sampling strategy, where samples are taken from every context, is termed a “blanket strategy.” A judgmental sampling strategy, in contrast, relies on the excavators making decisions about which contexts to sample (and which not to sample) during excavation. While this approach was used in the past, with excavators targeting archaeological features they thought would be more likely to include paleoethnobotanical remains, most archaeologists today acknowledge that it is difficult to anticipate where plants were used and disposed of within a site. As a result, most sampling today is blanket sampling. Some areas, such as the floors of rooms, might be further sampled, with a soil sample taken every one or two meters across the surface in a grid.

TR 29 - Topplan



An example of a sampling strategy in which soil samples, for both botanical macroremains and phytolith analysis, were taken every one meter in a hexagonal grid across the floor of two

rooms, one of which appears to have been used for food storage, and the other, for cooking. Note that samples were also taken outside these rooms in the exterior courtyard; the room on the left was still under excavation, and its floor had not yet been reached.

Source: Dr. Marston

Processing Samples for Botanical Macroremains

Soil samples taken for macrobotanical analysis are most commonly processed through the method known as **flotation**. Flotation uses water to separate soil, rocks, and light particles that float, which include most carbonized plant remains. Such samples are usually large, 10 to 30 liters in volume (1 to 3 large buckets full), although occasionally as small as 1 liter each.

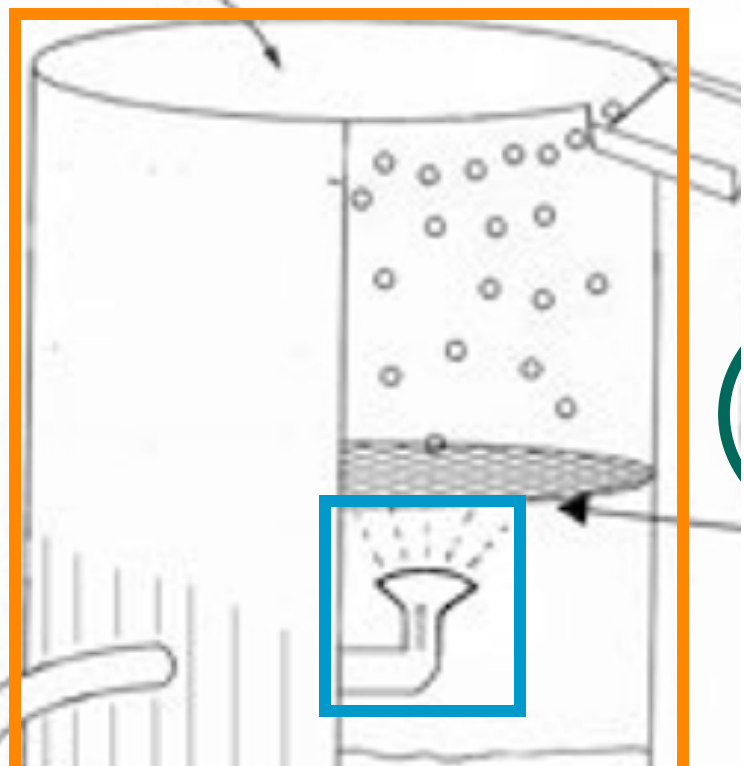
Flotation is a process that uses water to separate materials in a soil sample by density. The soil is dissolved and washed away, while heavy artifacts sink and are later dried and sorted. Most importantly, carbonized plant macroremains float and can be collected from the surface of the flotation tank.

The best way to understand the flotation process is to see it in action. Simplest is a method of bucket flotation, such as that shown here by Dr. Karen Adams.



To handle larger samples and more of them, a motorized flotation system can be employed. Peruse a diagram of such a machine below and then watch the video of Prof. Marston working with students in Hungary.

Click on each part of





Processing Samples for Botanical Microremains

Microremains are more difficult to extract. Paleoethnobotanists need to remove the soil particles and other materials from the botanical samples of interest. A variety of techniques for doing this exist, but all involve a series of chemical and mechanical processes that separate particles based on solubility and density. This work requires a laboratory with a chemical-safety fume hood, as powerful acids and other toxic substances are often used. Different protocols are best for each category of botanical microremains, although there is a method that can recover all three types (phytoliths, pollen, and starch granules) at once.

When an artifact is sampled for phytoliths and/or starch granules, it is a much simpler procedure. The artifact is simply washed with deionized water, and the dirty water running off the artifact is collected for microscopic analysis, either with or without chemical processing first. This is often done with careful agitation of small amounts of water against the artifact to dislodge starch and phytoliths, as depicted here:



Dr. Stephanie Simms, a BU Ph.D. alumna (GRS '14), sampling a mano (grinding stone) from a Maya site in Mexico for phytoliths and starch granules.

Source: Dr. Marston